

Harmonic Canvas: Inversion-Free Editing for Visually-Guided Music Style Transfer

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Abstract

Music style transfer (MST) aims to reinterpret existing musical pieces in new stylistic forms while maintaining their melodic coherence. Conventional approaches conditioned on text or audio overlook the profoundly multimodal character of musical style. Visual ambience – reflected in color, lighting, and composition – encodes affective attributes that parallel timbre, rhythm, and harmony, which, however, remain underexplored in MST context. We introduce a flow-based, inversion-free framework for multimodal music style transfer that unifies textual and visual guidance. Our approach tackles two challenges: (1) capturing cross-modal semantics beyond language through a dual-encoder fusion module that merges CLIP- and ViT-derived embeddings, and (2) preserving melodic identity using a differentiable normalized chroma constraint that regulates pitch-class consistency along the generative flow. We reorganize and extend the MeLBench and MusicCaps collections into a genre-structured multimodal dataset to support style-aware analysis. Quantitative and perceptual evaluations demonstrate that our approach achieves superior control, structural fidelity, and cross-modal expressiveness, underscoring the role of visual perception in music generation. Our demo is available [online](#).

1. Introduction

Music stands at the convergence of art, technology, and culture, underpinning modern creative industries that span short-form social media, film scoring, and interactive entertainment. It functions not only as an expressive medium but also as a cultural bridge that connects people through shared emotion and rhythm. With the rapid progress of generative modeling, music creation has become one of the most dynamic areas of computational creativity [54]. Deep genera-

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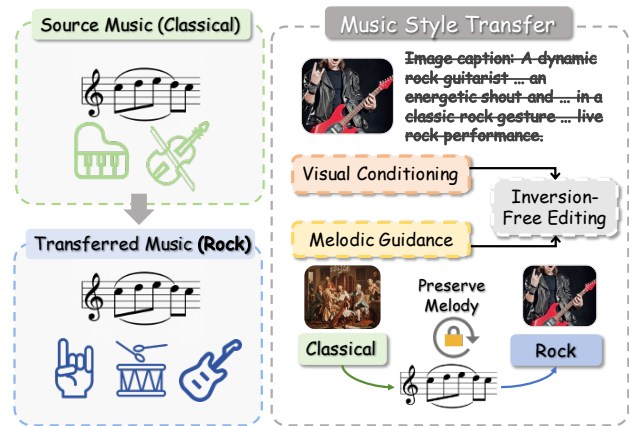


Figure 1. Overview of our Music Style Transfer workflow. Our model transforms a source clip (e.g., classical) into a target style (e.g., rock) via multimodal guidance. While the framework natively supports textual captions (strikethrough in figure to emphasize independence), it could leverages visual cues for non-verbal style and melodic guidance for pitch preservation.

tive approaches such as autoregressive models [55], generative adversarial network-based models [12] and diffusion models [15] have achieved remarkable success in music composition [8, 29, 32], enabling the creation of coherent melodies, realistic timbres, and stylistically rich arrangements from scratch.

While music generation from scratch constitutes a “zero-to-one” creative breakthrough, music style transfer (MST) – the transformation of existing compositions into new stylistic forms – represents the “one-to-one-hundred” stage, emphasizing scalability and creative accessibility. MST enables users to reinterpret familiar pieces through novel aesthetic perspectives without requiring expert compositional skills. Recent studies have approached this problem using diverse architectures, including spectrogram-based translation [59], diffusion-based editing frameworks [27] and extended to latent diffusion [33]. However, these methods pre-

dominantly depend on textual prompts or reference audio, leaving the multimodal nature of musical style – its interplay with visual and contextual cues – largely unexplored.

In particular, the visual modality remains a largely untapped resource for guiding musical style, despite its strong psychological and aesthetic connections to sound [42, 64]. Visual imagery, much like music, tells an affective story – it conveys atmosphere, mood, and emotional tone beyond the limits of language, embodying the very essence of what we perceive as style [3]. Visual cues such as *color* and *lighting* often map intuitively to musical attributes like *timbre*, *rhythm*, and *harmony*, suggesting that style perception is not purely auditory but intrinsically multimodal. Composers and producers frequently describe mood or intent through both visual references (e.g., *impressionist paintings*, *retro album covers*) and textual guidance (e.g., “*make it more melancholic, like a rainy afternoon*”), reflecting this cross-modal interplay between imagery and sound.

Despite these cross-modal correspondences, current music style transfer (MST) systems seldom exploit visual cues explicitly. Most rely on textual prompts or reference audio [27, 34], with text functioning as a compact – but inherently lossy – surrogate for visual ambience. While captions can approximate moods (e.g., “*bright and cheerful jazz*”), they fail to capture finer stylistic nuances such as color tone, spatial composition, or emotional texture – qualities that are more naturally expressed through visual media. This limitation motivates our first research challenge: *how can visual semantics that transcend linguistic description be effectively represented and utilized for style transfer?*

Meanwhile, advances in inversion-free editing [23, 60] offer a promising complementary direction. Unlike diffusion-based inversion, which introduces stochastic noise during reconstruction, inversion-free flow models establish deterministic transport paths between source and target distributions. These continuous mappings preserve global coherence while allowing localized stylistic variation – an essential property for music, where melodic and rhythmic integrity should be maintained. However, even within this structure-preserving formulation, abrupt stylistic shifts may still distort melodic patterns. This exposes our second challenge: *how to preserve pitch and rhythmic identity under large stylistic transformations within an inversion-free generative framework?*

To address these challenges, we propose a unified multimodal MST framework grounded in inversion-free flow editing [23], as illustrated in Fig. 1. First, we introduce a cross-modal fusion module that directly integrates visual embeddings with textual semantics, enabling perception and translation of non-verbal stylistic cues – such as ambience, tone, and spatial mood – into coherent musical expression. Second, to stabilize melodic structure, we incorporate a normalized chroma-based constraint that pre-

serves pitch-class consistency during generation, ensuring structural fidelity while allowing flexible stylistic adaptation. In addition, we construct a new multimodal dataset of aligned triplets (*music*, *text*, and *image*) to facilitate systematic studies of cross-modal conditioning in music style transfer. In summary, our main contributions are threefold:

- **A multimodal MST task.** We introduce the multimodal music style transfer (MST) task, which promotes a cross-modal cognitive perspective by aligning musical, visual, and linguistic modalities within a coherent representational paradigm for MST.
- **Non-verbal stylistic representation.** We propose a cross-modal fusion module that integrates visual embeddings with textual semantics, effectively capturing non-verbal ambience cues, such as color, tone, and spatial mood, that are critical to stylistic perception.
- **Inversion-free flow with melodic regularization.** We develop an inversion-free flow-based editing approach equipped with a normalized chroma constraint, achieving structure-preserving and melody-consistent style transformations.
- **A new MST benchmark.** We reorganize and extend MeLBench and MusicCaps into a more structured multimodal corpus with refined captions and 16-genre annotations, establishing a new benchmark for style-aware and multimodal MST evaluation.

2. Related Work

Generative-Music Creation. Music, as a uniquely expressive art form, has long attracted research interest in its computational creation methods. Early attempts at music generation relied on symbolic rules and probabilistic models such as Markov chains and HMMs [16, 47, 58] could capture simple melodic patterns but lacked expressive control and scalability. With the advent of deep learning, generative models based on VAEs and GANs greatly advanced the field. MusicVAE [48] introduced a hierarchical structure to model long-term dependencies; MuseGAN [8] and MidiNet [62] demonstrated GAN-based multi-track generation. More recently, diffusion models [18, 29, 32, 35] have achieved state-of-the-art fidelity by iteratively refining noise under semantic or textual guidance, offering strong controllability and realism. Beyond text-only conditioning, multimodal approaches such as MelFusion [3], MusFlow [53], and Diff-BGM [26] incorporate visual or cross-modal cues to enhance stylistic expressiveness in music generation. However, these models primarily focus on unconditional or from-scratch synthesis and lack mechanisms to preserve melodic and rhythmic content – key to effective music style transfer.

Music Style Transfer. Music style transfer (MST) has gradually evolved into an important branch of generative music research, aiming to re-render existing pieces

in new stylistic forms while retaining their musical identity. While earlier studies explored various generative deep learning paradigms for this task [20, 39], recent progress in diffusion-based generation [44, 61] has opened new possibilities for high-fidelity and controllable music stylization. Recent progress in text-conditioned music generation has further advanced controllable stylization. MuLan [17] introduces a joint audio–text embedding that supports text-guided re-rendering of a given excerpt. Noise2Music [18] extends this idea by using a cascade of text-conditioned diffusion models that can be applied to modify an input clip at high fidelity. Meanwhile, large-scale language–audio models such as MusicGen [5] and MusicLM [1] further extend style music generation by conditioning the process on both textual prompts and optional melodic inputs. In parallel, diffusion-based instruction-driven frameworks [14, 57] have been explored for general audio or music editing, where operations such as Add, Remove, or Replace are learned from supervised (prompt, input, output) triplets.

Inversion-based Editing Methods. Beyond generative music models trained from scratch using text or instructions, recent studies have explored leveraging pretrained diffusion priors for unsupervised style transfer and zero-shot editing. This direction was first established in the image domain, where methods such as SDEdit [36] and Pix2Pix-Zero [43] demonstrated that perturbations in noise space can enable semantic edits while preserving structural fidelity. These methods rely on diffusion inversion [19, 37] to reconstruct latent noise trajectories from real samples and enable semantic manipulation.

Building on these developments, diffusion inversion has been adapted to the audio domain for tasks such as timbre conversion and music resynthesis [27, 30, 34], showing that pretrained diffusion priors can support zero-shot or text-driven music editing. However, inversion requires stochastic multi-step reconstruction, making it computationally expensive and susceptible to accumulated errors [25] that cause temporal drift in long-form audio. To alleviate these limitations, inversion-free flow-based frameworks [23, 60] learn source-to-target transformations directly in latent space, offering a more stable and efficient alternative. While such formulations have shown strong performance in visual domains, their potential for structured musical transformation remains underexplored. In this work, we extend the inversion-free flow paradigm to music style transfer, introducing melody-preserving constraints and multimodal conditioning for rhythm-consistent and visually guided musical editing.

3. Preliminary

Music Style Transfer. Music style transfer (MST) aims to generate a stylized audio x_{sty} from a source clip x_{src} , preserving melody and rhythm while altering stylistic char-

acteristics [13]. Traditional MST relies on a single reference or domain target, whereas our extended formulation introduces multimodal style conditions (T, I) , where text T provides explicit intent and image I conveys complementary visual ambience. This multimodal conditioning enables finer control and more expressive style reinterpretation.

Inversion-Free Flow Editing. Flow-based generative models [28, 31] define data synthesis as a continuous transformation between distributions:

$$\frac{dz_t}{dt} = v_\theta(z_t, t), \quad z_0 \sim p_0, \quad z_1 \sim p_1, \quad (1)$$

where v_θ is a learnable velocity field driving the flow from the base distribution p_0 (e.g., Gaussian noise) to the data distribution p_1 . Unlike diffusion models that approximate a stochastic reverse process, flow matching defines a learning objective that aligns the model-predicted velocity with the target flow:

$$\mathcal{L}_{\text{flow}} = \mathbb{E}_{t, z_t} [\|v_\theta(z_t, t) - v_t\|_2^2], \quad (2)$$

where v_t represents the target flow velocity derived from interpolated data pairs. This formulation encourages path consistency and smooth transitions in latent space, allowing each intermediate state z_t to correspond to a continuous transformation of the data distribution. To enable efficient editing, inversion-free flow methods [23] bypass explicit noise reconstruction by directly constructing a mapping between source and target trajectories:

$$z_t^{\text{edit}} = z_t^{\text{src}} + \Delta v_t; \quad \Delta v_t = v_\theta(z_t^{\text{tar}}, t) - v_\theta(z_t^{\text{src}}, t), \quad (3)$$

thus realizing semantic transformation without diffusion inversion. This design eliminates stochastic noise accumulation and multi-step inversion, providing a stable and directionally consistent mapping ideal for structure-preserving tasks such as music style transfer. In our framework, we build upon this inversion-free flow formulation to achieve efficient and multimodally guided music editing.

4. Methodology

In this section, we introduce a multimodal fusion module that integrates visual and textual cues into the audio backbone (Section 4.1), followed by an inversion-free editing framework with chroma guidance (Section 4.2) to preserve melodic and rhythmic structure. Together, they form a unified pipeline for multimodal music style transfer (Fig. 2).

4.1. Cross-Adapter for Visual-Textual Conditioning

Style information in music is often conveyed not only by sound but also by visual ambience – color palettes, lighting, composition, and texture – that language alone cannot fully describe. This raises a natural question: how can image information be effectively integrated into the model?

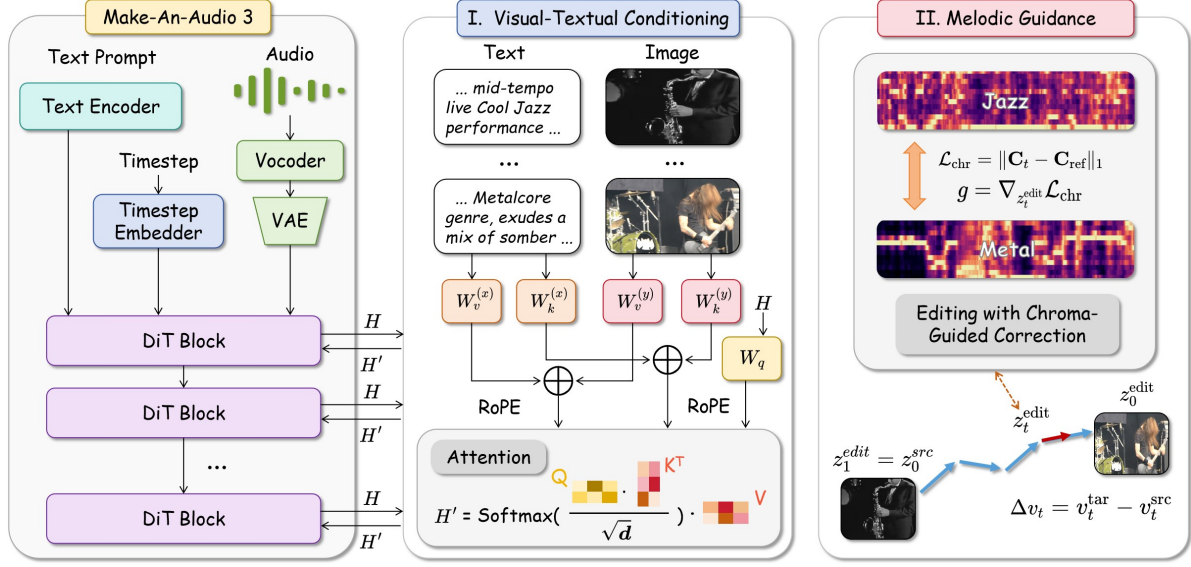


Figure 2. Overview of the proposed multimodal music style transfer framework. (Left) We build upon the Make-An-Audio 3 backbone and perform inversion-free flow editing in latent space. (Middle) Visual Conditioning: text and image features are encoded separately and injected into each DiT block via a cross-adapter attention mechanism, enabling the model to interpret non-verbal aesthetics such as ambience, color tone, and scene mood. (Right) Melodic Guidance: a normalized chroma loss constrains pitch-class structure, and its gradient corrects the flow trajectory, ensuring that stylistic transformation preserves melodic identity.

A straightforward solution is to use existing text-to-audio models by converting image descriptions into text. However, from an information-theoretic standpoint [52], this introduces information loss. Let I be the reference image, $C = g(I)$ its caption, and Z the generated audio. Since $I \rightarrow C \rightarrow Z$ is a Markov chain, the data-processing inequality gives $\mathcal{I}(I; Z) \leq \mathcal{I}(I; C)$, meaning that the caption C can only lose, not gain, stylistic information from I . Hence, conditioning on visual features provides richer style cues than using image captions alone. Empirically, our experiments indicate that captioning an image often fails to capture fine-grained aesthetic cues relevant to musical style. Motivated by this, we incorporate visual features alongside textual prompts, allowing the model to perceive and translate musical style in a more holistic and expressive manner.

To support such multimodal conditioning, a flexible and efficient generative backbone is essential. We adopt Make-An-Audio 3 (MAA3) [65], an enhanced transformer-based diffusion architecture that extends DiT with improved normalization, frequency-aware attention, and adaptive modulation for stronger conditional control. Unlike convolutional backbones, MAA3 tokenizes latent features, enabling efficient modeling of long-range temporal dependencies—an essential property for music. In the music generation setup, audio waveforms are first converted into Mel-spectrograms via STFT and encoded by a lightweight audio VAE into latent tokens. A pretrained Flan-T5 text encoder [4] provides semantic conditioning through adaptive layer normalization

(AdaLN) in each transformer block, while rotary positional embeddings (RoPE) capture temporal structure and maintain cross-layer alignment. This token-based backbone offers a flexible foundation for incorporating additional conditional streams, making it particularly suitable for our proposed cross-adapter integration of visual information.

Building upon this foundation, we introduce a cross-adapter mechanism to integrate visual features into the transformer without disrupting its pretrained structure. Specifically, we employ two complementary visual encoders – CLIP [45] and ViT [9] – to capture distinct aspects of visual style. CLIP excels at learning high-level semantic correspondences between vision and language, providing conceptually aligned embeddings, whereas ViT focuses on global information [46]. The resulting embeddings are pooled, linearly projected, and injected into the transformer via a cross-attention mechanism, enabling the generator to translate rich non-verbal cues – such as color tone, lighting, and mood – into coherent musical expression. For each transformer block, the main stream and control stream share the query projection but maintain distinct key and value projections, formulated as:

$$\begin{cases} Q_m = W_q \text{Norm}(H) \\ K_{\text{ctl}} = W_k^{(x)} \Phi_x(X_{\text{ctl}}) + W_k^{(y)} \Phi_y(Y_{\text{ctl}}) \\ V_{\text{ctl}} = W_v^{(x)} \Phi_x(X_{\text{ctl}}) + W_v^{(y)} \Phi_y(Y_{\text{ctl}}) \end{cases} \quad (4)$$

where H is the hidden representation of the main stream, X_{ctl} and Y_{ctl} denote control-side textual and visual features,

Algorithm 1 Inversion-free flow editing with chroma-guided correction

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1: Input: Source latent  $x_{\text{src}}$ , noise  $x_0^{\text{noise}}$ , timesteps  $\{t_i\}_{i=0}^{T-1}$ , decoder  $\mathcal{G}$ , chroma extractor  $\Phi_{\text{chroma}}$ , inner steps  $S$ , guidance rate  $\eta$ , weight  $\lambda_{\text{chr}}$ 
2: Output: Final edited latent  $z_{t_{T-1}}^{\text{edit}}$ 
3: Init:  $z_{t_0}^{\text{edit}} = x_{\text{src}}$ 
4: for  $i = 0$  to  $T - 2$  do
5:    $dt = t_{i+1} - t_i$ 
6:    $z_{t_i}^{\text{src}} = (1 - (1 - \sigma_{\min})t_i)x_{\text{src}} + t_ix_0^{\text{noise}}$ 
7:    $\Delta v = f_{\theta}(z_{t_i}^{\text{tar}}, y_{\text{tar}}) - f_{\theta}(z_{t_i}^{\text{src}}, y_{\text{src}})$ 
8:    $z_{t_i}^{\text{edit}} = z_{t_i}^{\text{edit}} + dt v_{\Delta}$ 
9:   for  $s = 1$  to  $S$  do
10:     $\mathbf{C}_{t_i}, \mathbf{C}_{\text{ref}} = \Phi_{\text{chroma}}(\mathcal{G}(z_{t_i}^{\text{edit}})), \Phi_{\text{chroma}}(\mathcal{G}(z_{t_i}^{\text{src}}))$ 
11:     $\mathcal{L}_{\text{chr}} = \|\mathbf{C}_{t_i} - \mathbf{C}_{\text{ref}}\|_1$ 
12:     $g = \nabla_{z_{t_i}^{\text{edit}}} \mathcal{L}_{\text{chr}}$ 
13:     $z_{t_i}^{\text{edit}} = z_{t_i}^{\text{edit}} - \eta \lambda_{\text{chr}} g$ 
14:   end for
15:    $z_{t_{i+1}}^{\text{edit}} = z_{t_i}^{\text{edit}}$ 
16: end for
17: return  $z_{t_{T-1}}^{\text{edit}}$ 

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W_q, W_k, W_v are the projection matrices for query, key, and value, and $\Phi_x(\cdot), \Phi_y(\cdot)$ are linear transformations applied to each modality before fusion. The cross-adapter attention output is then computed as:

$$A_{\text{crs}} = \text{softmax}\left(\frac{(Q_m \odot R_q)(K_{\text{ctl}} \odot R_k)^{\top}}{\sqrt{d}}\right), \quad (5)$$

$$O_{\text{crs}} = A_{\text{crs}} V_{\text{ctl}} W_o, \quad (6)$$

where d represents the head dimension, Φ_x, Φ_y are pre-normalization mappings, and R_q, R_k denote segmented RoPE factors with separate time grids for main and control streams. We share W_q and W_o across both streams to keep their representation spaces aligned, while keeping $W_k^{(x)}, W_v^{(x)}, W_k^{(y)}, W_v^{(y)}$ distinct for the control path. Key normalization is applied for numerical stability. Then the adapter output is added to the main stream.

During training, most control-stream parameters that do not contribute to the main loss – such as feed-forward and normalization layers on the control side – are kept frozen to avoid unused-parameter issues and ensure stable convergence. This cross-adapter architecture allows the main stream to query the visual branch at each layer, injecting ambience and mood cues while preserving temporal and melodic structure. Together, these components enable direct control of musical style from textual and visual inputs, forming the foundation of our multimodal MST framework.

4.2. Melody Preservation via Melodic Guidance

While multimodal conditioning enriches stylistic expressiveness, it may also disrupt melodic stability. In practice, we observe that when the model relies solely on semantic cues from text or images, the generated audio can exhibit pitch drift or rhythmic distortion – altering the musical identity of the source. To mitigate this issue, we introduce a melody guidance mechanism that explicitly constrains the pitch-class distribution of the generated signal into the inversion-free flow framework. We illustrate the algorithmic details in Algorithm 1.

Our approach leverages the normalized chroma representation [2], a compact descriptor of melody that captures the relative energy distribution across the twelve pitch classes within an octave. Compared to raw fundamental frequency (F_0) curves, normalized chroma is amplitude-invariant, timbre-robust, and polyphony-tolerant, providing a more stable basis for constraining melodic consistency. Specifically, given the current waveform reconstruction $\hat{x}_t = \mathcal{G}(z_t^{\text{edit}})$ decoded from the edited latent state z_t^{edit} , we extract its chroma features as $\mathbf{C}_t = \Phi_{\text{chroma}}(\hat{x}_t)$, and compare them against a reference chroma $\mathbf{C}_{\text{ref}} = \Phi_{\text{chroma}}(x_{\text{ref}})$ computed from the instrumental stem of the source audio. The melodic deviation is measured via an L_1 loss:

$$\mathcal{L}_{\text{chr}} = \|\mathbf{C}_t - \mathbf{C}_{\text{ref}}\|_1, \quad (7)$$

which penalizes deviations in pitch-class activation while tolerating small timing shifts. The gradient of this loss with respect to the latent state is

$$g = \nabla_{z_t^{\text{edit}}} \mathcal{L}_{\text{chr}}(z_t^{\text{edit}}), \quad (8)$$

where g is the chroma-guidance direction in latent space. The latent is then refined through a proximal update:

$$z_t^{\text{edit}} \leftarrow z_t^{\text{edit}} - \eta \lambda_{\text{chr}} g, \quad (9)$$

where $\eta > 0$ is the inner-step learning rate and $\lambda_{\text{chr}} > 0$ controls the strength of melodic guidance. This update is repeated for S inner iterations at each flow step t , so that the outer inversion-free flow provides semantic style transformation, while the inner chroma-guided corrections pull the trajectory back toward the source pitch-class structure. In practice, we employ a cosine-decayed schedule for λ_{chr} — strong in early steps to “lock” melody, and weaker in later steps to release stylistic flexibility. The chroma extractor Φ_{chroma} is implemented with a differentiable Short-Time Fourier Transform followed by a 12-bin triangular filter bank and per-frame L_1 normalization, ensuring full gradient flow to the latent variables.

By integrating this normalized chroma constraint, the model maintains coherent pitch and rhythm while still performing expressive multimodal edits. This melody-guided

Table 1. Objective and subjective evaluation results. Two modality indicators (Text/Image) are shown for each method. Bold number denotes the best metric value.

Method	Modality		Objective Metrics					Subjective Metrics		
	Text	Image	FAD↓	FD↓	IMSM↑	F ₀ -PCC↑	CCS↑	OVL↑	REL↑	MOS _{con} ↑
MusicLM [1]	✓	✗	5.26	36.55	-	-	-	80.73	79.60	80.33
MusicGen [5]	✓	✗	4.89	34.81	-	-	-	81.60	80.40	81.13
TANGO [32]	✓	✗	4.59	29.96	-	-	-	83.27	82.33	82.93
MAA3 [65]	✓	✗	4.23	29.77	-	-	-	86.93	85.93	86.47
MeLFusion [3]	✓	✓	3.24	25.73	0.783	-	-	87.40	86.27	86.93
MusicTI [27]	✓	✗	3.78	26.09	-	0.367	0.820	86.80	85.47	86.13
ZETA [34]	✓	✗	4.23	27.75	-	0.322	0.764	84.53	83.27	83.93
Ours	✓	✓	2.43	24.06	0.828	0.416	0.878	89.27	88.13	89.20

mechanism thus enforces a meaningful regularization on the generative flow, achieving style transfer that preserves musical identity without sacrificing stylistic diversity.

5. Experiments and Results

5.1. Experiments Setup

Datasets. We construct a unified multimodal dataset by merging extended versions of MeLBench [3] and MusicCaps [1], both normalized into triplets $\langle I, T, M \rangle$ representing image, text, and music. Each triplet contains a visually descriptive image I , a caption T describing the musical intent, and an audio clip M aligned in style and emotion. To enhance stylistic consistency, we apply the stem-separation model Demucs [6] to remove vocals and retain purely instrumental tracks across both datasets. The MusicCaps subset is further extended with visually relevant images and enriched using an LLM-assisted annotation pipeline, which classifies each clip into one of 16 musical genres (e.g., ambient, jazz-fusion, electronic, retro-pop) and standardizes captions for stylistic coherence, additional details are provided in the Appendix. The final corpus provides genre-aware, visually grounded triplets with richer metadata, enabling fine-grained multimodal conditioning and style-transfer evaluation.

Evaluation Metrics. We evaluate our framework using both objective and subjective metrics, with detailed procedures and rating criteria provided in the Appendix. We evaluate the transferred music in terms of *content preservation* and *style similarity*. The main metrics include Fréchet Audio Distance (FAD) [21], Fréchet Distance (FD) [22], Image–Music Similarity Metric (IMSM) [3], Fundamental Frequency Correlation (F₀-PCC) [41], and Chroma-CQT Similarity (CCS) [51]. FAD and FD measure distributional alignment between generated and target-style audio in perceptual embedding spaces, while IMSM assesses cross-

modal consistency with visual–textual prompts. F₀-PCC and CCS capture melodic and harmonic preservation, reflecting how well pitch and tonal progression are maintained. Lower FAD/FD and higher IMSM, F₀-PCC, and CCS indicate better style transfer quality.

We conduct a user study to evaluate the perceptual quality of the transferred music. Following prior works in text-to-audio generation [3, 11, 29], three subjective measures are adopted: overall audio quality (OVL), prompt relevance (REL), and melodic consistency (MOS_{con}). Fifteen participants rated each sample on a 0–100 scale, where higher scores indicate better perceptual performance in terms of realism, stylistic relevance, and melody preservation.

Implementation Details. For fair comparison, we adopt MAA3 [65] as our backbone. The model is trained on L40S GPUs using the 1.5B backbone with an additional 46.2M adapter parameters. All baselines are evaluated under identical settings. The cross-adapter is trained with a learning rate of 1×10^{-5} and a batch size of 16. The melody-guided rate η is set to 1×10^{-4} with a step size of 2 and a weight λ_{chr} of 1. The inference step count is set to 25 following [65].

5.2. Baseline Models

We evaluate our method against representative baselines across both text-to-music generation and style-transfer settings. To the best of our knowledge, there is currently no diffusion-based music style transfer framework conditioned on both visual and textual modalities, making direct multimodal comparison unavailable. Therefore, we select state-of-the-art models according to their relevance in either text-conditioned generation or audio-based style transfer.

Generation-based Style Transfer. We include MusicLM [1], MusicGen [5], MAA3 [65], Tango [32], and MeLFusion [3], which represent recent advances in controllable music generation. While numerous generation systems

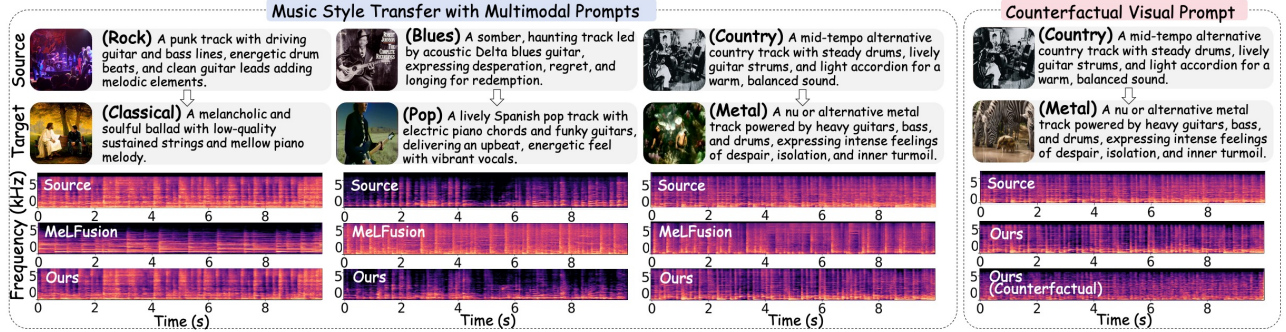


Figure 3. (Left) Visualization of music style transfer results across three multimodal prompts. For each prompt, source/target images are shown on the left, followed by Mel-spectrograms of the Source, MeLFusion, and Ours outputs. (Right) Counterfactual test: changing only the image while keeping text fixed leads to different outputs, demonstrating the influence of visual cues on generated musical style.

exist, such as Riffusion [10], Mubert [38], Moûsai [50], Noise2Music [18], MeLoDy [24], DiffRhythm [40], YuE [63], and SongComposer [7], they do not support the direct transformation of input audio or rely on specialized pipelines that hinder fair comparison. We thus report results for publicly available and widely adopted open-source models that reflect prevailing architectural trends. These baselines serve as upper-bound references for text-conditioned generation quality and style controllability.

Inversion-based Style Transfer. To assess content-preserving transformation, we further compare with ZETA [34] and MusicTI [27], two representative inversion-based frameworks explicitly designed for music style transfer. These systems aim to modify musical timbre or mood while maintaining melodic and rhythmic identity, aligning closely with our task formulation.

5.3. Overall Performance

We provide both objective and perceptual comparisons against representative baselines in Table 1. Compared with text-to-music generation systems such as MusicLM [1], MusicGen [5], and TANGO [32], our method performs a direct transformation rather than synthesis from scratch. These generative models produce coherent audio but lack access to source material, making structure-based metrics (e.g., F_0 -PCC, CCS) inapplicable; In contrast, our inversion-free flow editing preserves temporal and harmonic structures while enabling fine-grained style adaptation under multimodal control. Compared with MusicTI [27] and ZETA [34], our method achieves higher F_0 -PCC and CCS scores, possibly due to the chroma-based constraint, which helps maintain pitch stability and tonal coherence. Among multimodal approaches, only our method and MeLFusion [3] incorporate visual guidance. While MeLFusion benefits from image features, our cross-adaptor fusion achieves stronger alignment between visual ambience and musical expression, as reflected in the higher

IMSM score.

Additionally, the left panel of Fig. 3 compares Mel-spectrograms across multimodal prompts. A ideal music style transfer method should keep the source’s structural patterns (melodic contours, rhythmic events) while modifying only style-related energy distributions. Our method achieves this behavior, whereas MeLFusion exhibits noticeable structural deformation.

5.4. Ablation Study

Effect of the Image Modality. We assess the contribution of visual information through modality ablation and a counterfactual test. As shown in Table 2, text-only conditioning yields coherent transfers but lacks fine-grained stylistic cues. BLIP-generated captions slightly improve performance but remain inferior to direct visual embeddings, while image-only conditioning performs the worst, indicating that visual cues require textual grounding. Combining text and image achieves the best performance – especially in IMSM – showing that visual features provide complementary style information beyond text alone. We further observe that adding BLIP captions to text+image can degrade performance, likely due to low-quality or verbose captions introducing noise into the conditioning space. Furthermore, the counterfactual test (Fig. 3, right) shows that altering only the image while keeping the text fixed leads to clearly different outputs, demonstrating that visual cues independently affect the generated style.

Comparison of Inversion Strategies. Since our method is built upon an inversion-free flow formulation, we first compare it with FlowEdit [23], which also learns deterministic source–target mappings without reconstructing noise trajectories. For completeness, we further include RF Edit [56] and RF Inversion [49], both relying on explicit inversion in flow-based formulation. As shown in Tab. 3, our method achieves competitive overall performance and substantially higher F_0 -PCC and CCS. This comparison

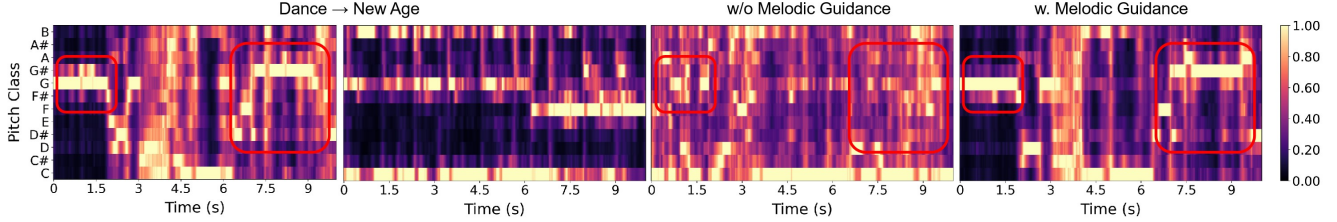


Figure 4. Comparison of normalized chroma energy curves across four settings: (a) source audio, (b) target style reference, (c) transfer without melody constraint, and (d) our chroma-guided method. The chroma distributions clearly illustrate how our approach maintains melodic consistency while achieving stylistic transformation.

Table 2. Ablation on modality conditioning. Checkmarks indicate which modalities are provided; *Caption* denotes a BLIP-generated caption from the image.

Setting	Modality			Metrics		
	Text	Image	Caption	FAD↓	FD↓	IMSM↑
Text only	✓	✗	✗	3.15	25.38	-
Image only	✗	✓	✗	4.06	34.13	-
Text + Image (Caption)	✓	✗	✓	3.49	31.24	-
Text + Image + Caption	✓	✓	✓	2.86	24.57	0.811
Text + Image	✓	✓	✗	2.43	24.06	0.828

Table 3. Comparison of Inversion Strategies. Lower is better for FAD/FD; higher is better for others.

Method	Melodic Consistency		Stylistic Consistency		
	F ₀ -PCC↑	CCS ↑	FAD↓	FD↓	IMSM↑
RF Inversion [49]	0.323	0.830	2.57	26.86	0.812
RF Edit [56]	0.343	0.842	2.52	26.18	0.805
FlowEdit [23]	0.368	0.852	2.41	24.45	0.829
Ours	0.416	0.878	2.43	24.06	0.828

highlights the advantage of inversion-free flow formulations over inversion-based schemes in preserving structural fidelity during style transfer.

Impact of Melodic Guidance. To isolate the effect of melodic guidance itself, we visualize the chroma maps shown in Fig. 4. In the transfer from *Dance* to *New Age*, guided editing preserves key melodic trajectories (red regions), whereas unguided editing exhibits noticeable chroma drift. This demonstrates that the chroma constraint stabilizes harmonic progression and helps maintain musical identity during style transfer. Moreover, the last two rows of Tab. 3 show that although introducing melodic guidance substantially improves the F₀-PCC and CCS scores, stylistic consistency decreases slightly. We attribute this to the inherent trade-off between preserving source-domain melodic structure and fully aligning with the target distribution.

Sensitivity Analysis. We analyze the sensitivity of chroma-guided flow editing to the guidance rate (GR) and the number of correction steps. Since each correction step operates purely through gradient updates on the latent state – without

Table 4. Sensitivity analysis of the guidance rate (GR) and the number of chroma-guided correction steps.

Varying Steps				Varying Guidance Rates			
GR	Steps	CCS↑	IMSM↑	Steps	GR	CCS↑	IMSM↑
1e−4	0	0.852	0.829	2	1e−5	0.869	0.828
	1	0.871	0.828		2e−5	0.873	0.828
	2	0.878	0.828		5e−5	0.875	0.828
	5	0.878	0.824		1e−4	0.878	0.828
	10	0.878	0.822		2e−4	0.877	0.827
	20	0.880	0.815		5e−4	0.876	0.825

additional neural network evaluations – the computational overhead is negligible; therefore, allowing us to focus exclusively on performance. As shown in Table 4, chroma guidance enhances melodic fidelity, yielding higher CCS scores, but slightly reduces IMSM due to stronger melodic anchoring. Increasing the number of correction steps leads to diminishing CCS gains while further lowering IMSM, suggesting that excessive correction limits stylistic flexibility. Similarly, larger guidance rates improve melody preservation up to a point, after which the constraint begins to dominate the latent flow and weaken style adaptation.

6. Conclusion

In this work, we introduced a multimodal, inversion-free flow framework for music style transfer that unifies visual and textual semantics within a coherent editing process. By combining a dual-encoder fusion module for cross-modal stylistic cues with a differentiable chroma-based constraint for melodic preservation, the model achieves style transformation while maintaining pitch-class structure and musical identity. Comprehensive objective and perceptual evaluations demonstrate consistent improvements in structural fidelity, stylistic controllability, and multimodal alignment. We further reorganized and extended MeLBench and MusicCaps into a genre-aware multimodal corpus, enabling more rigorous style-aware evaluation. Future directions include developing lighter flow architectures to pursue broader multimodal pretraining for more adaptive and interactive creative applications.

References

- [1] Andrea Agostinelli, Timo I Denk, Zalán Borsos, Jesse Engel, Mauro Verzett, Antoine Caillon, Qingqing Huang, Aren Jansen, Adam Roberts, Marco Tagliasacchi, et al. Musiclm: Generating music from text. *arXiv preprint arXiv:2301.11325*, 2023. 3, 6, 7
- [2] Mark A Bartsch and Gregory H Wakefield. Audio thumbnailing of popular music using chroma-based representations. *IEEE Transactions on Multimedia*, 7(1):96–104, 2005. 5
- [3] Sanjoy Chowdhury, Sayan Nag, KJ Joseph, Balaji Vasan Srinivasan, and Dinesh Manocha. Melfusion: Synthesizing music from image and language cues using diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26826–26835, 2024. 2, 6, 7
- [4] Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, et al. Scaling instruction-finetuned language models. *Journal of Machine Learning Research*, 25(70):1–53, 2024. 4
- [5] Jade Copet, Felix Kreuk, Itai Gat, Tal Remez, David Kant, Gabriel Synnaeve, Yossi Adi, and Alexandre Défossez. Simple and controllable music generation. *Advances in Neural Information Processing Systems*, 36:47704–47720, 2023. 3, 6, 7
- [6] Alexandre Défossez, Nicolas Usunier, Léon Bottou, and Francis Bach. Demucs: Deep extractor for music sources with extra unlabeled data remixed. *arXiv preprint arXiv:1909.01174*, 2019. 6
- [7] Shuangrui Ding, Zihan Liu, Xiaoyi Dong, Pan Zhang, Rui Qian, Junhao Huang, Conghui He, Dahua Lin, and Jiaqi Wang. Songcomposer: A large language model for lyric and melody generation in song composition. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7108–7127, 2025. 7
- [8] Hao-Wen Dong, Wen-Yi Hsiao, Li-Chia Yang, and Yi-Hsuan Yang. Musegan: Multi-track sequential generative adversarial networks for symbolic music generation and accompaniment. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018. 1, 2
- [9] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 4
- [10] S. Forsgren and H. Martiros. Riffusion - stable diffusion for real-time music generation, 2022. 7
- [11] Deepanway Ghosal, Navonil Majumder, Ambuj Mehrish, and Soujanya Poria. Text-to-audio generation using instruction guided latent diffusion model. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 3590–3598, 2023. 6
- [12] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial networks. *Communications of the ACM*, 63(11):139–144, 2020. 1
- [13] Eric Grinstein, Ngoc QK Duong, Alexey Ozerov, and Patrick Pérez. Audio style transfer. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 586–590. IEEE, 2018. 3
- [14] Bing Han, Junyu Dai, Weituo Hao, Xinyan He, Dong Guo, Jitong Chen, Yuxuan Wang, Yanmin Qian, and Xuchen Song. Instructme: an instruction guided music edit framework with latent diffusion models. In *Proceedings of the 33rd International Joint Conference on Artificial Intelligence*, pages 5835–5843, 2024. 3
- [15] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33:6840–6851, 2020. 1
- [16] Chih-Fang Huang, Yu-Shian Lian, Wei-Po Nien, and Wei-Hua Chieng. Analyzing the perception of chinese melodic imagery and its application to automated composition. *Multimedia Tools and Applications*, 75(13):7631–7654, 2016. 2
- [17] Qingqing Huang, Aren Jansen, Joonseok Lee, Ravi Ganti, Judith Yue Li, and Daniel PW Ellis. Mulan: A joint embedding of music audio and natural language. *arXiv preprint arXiv:2208.12415*, 2022. 3
- [18] Qingqing Huang, Daniel S Park, Tao Wang, Timo I Denk, Andy Ly, Nanxin Chen, Zhengdong Zhang, Zhishuai Zhang, Jiahui Yu, Christian Frank, et al. Noise2music: Text-conditioned music generation with diffusion models. *arXiv preprint arXiv:2302.03917*, 2023. 2, 3, 7
- [19] Inbar Huberman-Spiegelglas, Vladimir Kulikov, and Tomer Michaeli. An edit friendly ddpm noise space: Inversion and manipulations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12469–12478, 2024. 3
- [20] Junyan Jiang, Gus G Xia, Dave B Carlton, Chris N Anderson, and Ryan H Miyakawa. Transformer vae: A hierarchical model for structure-aware and interpretable music representation learning. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 516–520. IEEE, 2020. 3
- [21] Kevin Kilgour, Mauricio Zuluaga, Dominik Roblek, and Matthew Sharifi. Fréchet audio distance: A reference-free metric for evaluating music enhancement algorithms. In *Proc. Interspeech*, pages 2350–2354, 2019. 6
- [22] Qiuqiang Kong, Yin Cao, Turab Iqbal, Yuxuan Wang, Wenwu Wang, and Mark D Plumbley. Panns: Large-scale pretrained audio neural networks for audio pattern recognition. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 28:2880–2894, 2020. 6
- [23] Vladimir Kulikov, Matan Kleiner, Inbar Huberman-Spiegelglas, and Tomer Michaeli. Flowedit: Inversion-free text-based editing using pre-trained flow models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 19721–19730, 2025. 2, 3, 7, 8
- [24] Max WY Lam, Qiao Tian, Tang Li, Zongyu Yin, Siyuan Feng, Ming Tu, Yuliang Ji, Rui Xia, Mingbo Ma, Xuchen Song, et al. Efficient neural music generation. *Advances in Neural Information Processing Systems*, 36:17450–17463, 2023. 7
- [25] Ruibin Li, Ruihuang Li, Song Guo, and Lei Zhang. Source prompt disentangled inversion for boosting image editability

- with diffusion models. In *European Conference on Computer Vision*, pages 404–421. Springer, 2024. 3
- [26] Sizhe Li, Yiming Qin, Minghang Zheng, Xin Jin, and Yang Liu. Diff-bgm: A diffusion model for video background music generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27348–27357, 2024. 2
- [27] Sifei Li, Yuxin Zhang, Fan Tang, Chongyang Ma, Weiming Dong, and Changsheng Xu. Music style transfer with time-varying inversion of diffusion models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 547–555, 2024. 1, 2, 3, 6, 7
- [28] Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023. 3
- [29] Haohe Liu, Zehua Chen, Yi Yuan, Xinhao Mei, Xubo Liu, Danilo Mandic, Wenwu Wang, and Mark D Plumbley. Audioldm: Text-to-audio generation with latent diffusion models. In *International Conference on Machine Learning*, pages 21450–21474. PMLR, 2023. 1, 2, 6
- [30] Huadai Liu, Jialei Wang, Xiangtai Li, Rongjie Huang, Yang Liu, Jiayang Xu, and Zhou Zhao. Medic: Zero-shot music editing with disentangled inversion control. *arXiv preprint arXiv:2407.13220*, 2024. 3
- [31] Xingchao Liu, Chengyue Gong, et al. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations*, 2023. 3
- [32] Navonil Majumder, Chia-Yu Hung, Deepanway Ghosal, Wei-Ning Hsu, Rada Mihalcea, and Soujanya Poria. Tango 2: Aligning diffusion-based text-to-audio generations through direct preference optimization. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 564–572, 2024. 1, 2, 6, 7
- [33] Michele Mancusi, Yurii Halychanskyi, Kin Wai Cheuk, Eloi Moliner, Chieh-Hsin Lai, Stefan Uhlich, Junghyun Koo, Marco A Martínez-Ramírez, Wei-Hsiang Liao, Giorgio Fabbro, et al. Latent diffusion bridges for unsupervised musical audio timbre transfer. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE, 2025. 1
- [34] Hila Manor and Tomer Michaeli. Zero-shot unsupervised and text-based audio editing using ddpm inversion. In *International Conference on Machine Learning*, pages 34603–34629, 2024. 2, 3, 6, 7
- [35] Giorgio Mariani, Irene Tallini, Emilian Postolache, Michele Mancusi, Luca Cosmo, and Emanuele Rodolà. Multi-source diffusion models for simultaneous music generation and separation. In *International Conference on Learning Representations*, 2024. 2
- [36] Chenlin Meng, Yutong He, Yang Song, Jiaming Song, Jianjun Wu, Jun-Yan Zhu, and Stefano Ermon. Sdedit: Guided image synthesis and editing with stochastic differential equations. In *International Conference on Learning Representations*, 2022. 3
- [37] Ron Mokady, Amir Hertz, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Null-text inversion for editing real images using guided diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6038–6047, 2023. 3
- [38] Mubert-Inc. Mubert ai music generator — royalty free music, 2023. 7
- [39] Aashiq Muhamed, Liang Li, Xingjian Shi, Suri Yaddanapudi, Wayne Chi, Dylan Jackson, Rahul Suresh, Zachary C Lipton, and Alex J Smola. Symbolic music generation with transformer-gans. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 408–417, 2021. 3
- [40] Ziqian Ning, Huakang Chen, Yuepeng Jiang, Chunbo Hao, Guobin Ma, Shuai Wang, Jixun Yao, and Lei Xie. Diffrrhythm: Blazingly fast and embarrassingly simple end-to-end full-length song generation with latent diffusion. *arXiv preprint arXiv:2503.01183*, 2025. 7
- [41] Xinlei Niu, Jing Zhang, and Charles Patrick Martin. Hybridvc: Efficient voice style conversion with text and audio prompts. In *Proc. Interspeech*, pages 4368–4372, 2024. 6
- [42] Stephen E Palmer, Karen B Schloss, Zoe Xu, and Lilia R Prado-León. Music–color associations are mediated by emotion. *Proceedings of the National Academy of Sciences*, 110(22):8836–8841, 2013. 2
- [43] Gaurav Parmar, Krishna Kumar Singh, Richard Zhang, Yijun Li, Jingwan Lu, and Jun-Yan Zhu. Zero-shot image-to-image translation. In *ACM SIGGRAPH 2023 Conference Proceedings*, pages 1–11, 2023. 3
- [44] Vadim Popov, Ivan Vovk, Vladimir Gogoryan, Tasnima Sadekova, and Mikhail Kudinov. Grad-tts: A diffusion probabilistic model for text-to-speech. In *International Conference on Machine Learning*, pages 8599–8608. PMLR, 2021. 3
- [45] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pages 8748–8763. PMLR, 2021. 4
- [46] Maithra Raghu, Thomas Unterthiner, Simon Kornblith, Chiyuan Zhang, and Alexey Dosovitskiy. Do vision transformers see like convolutional neural networks? *Advances in Neural Information Processing Systems*, 34:12116–12128, 2021. 4
- [47] Adhika Sigit Ramanto and Nur Ulfa Maulidevi. Markov chain based procedural music generator with user chosen mood compatibility. *International Journal of Asia Digital Art and Design*, 21(1):19–24, 2017. 2
- [48] Adam Roberts, Jesse Engel, Colin Raffel, Curtis Hawthorne, and Douglas Eck. A hierarchical latent vector model for learning long-term structure in music. In *International Conference on Machine Learning*, pages 4364–4373. PMLR, 2018. 2
- [49] Litu Rout, Yujia Chen, Nataniel Ruiz, Constantine Caramanis, Sanjay Shakkottai, and Wen-Sheng Chu. Semantic image inversion and editing using rectified stochastic differential equations. In *International Conference on Learning Representations*, 2025. 7, 8

- [50] Flavio Schneider, Ojasv Kamal, Zhijing Jin, and Bernhard Schölkopf. Moûsai: Efficient text-to-music diffusion models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8050–8068, 2024. [7](#)
- [51] Christian Schörrhuber and Anssi Klapuri. Constant-q transform toolbox for music processing. In *7th sound and music computing conference, Barcelona, Spain*, pages 3–64. SMC, 2010. [6](#)
- [52] Claude E Shannon. A mathematical theory of communication. *The Bell system technical journal*, 27(3):379–423, 1948. [4](#)
- [53] Jiahao Song and Yu-Zhao Wang. Musflow: Multimodal music generation via conditional flow matching. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pages 10200–10209, 2025. [2](#)
- [54] Inc. Suno. Suno, 2023. AI music generation platform. [1](#)
- [55] Aäron Van Den Oord, Nal Kalchbrenner, and Koray Kavukcuoglu. Pixel recurrent neural networks. In *International Conference on Machine Learning*, pages 1747–1756. PMLR, 2016. [1](#)
- [56] Jiangshan Wang, Junfu Pu, Zhongang Qi, Jiayi Guo, Yue Ma, Nisha Huang, Yuxin Chen, Xiu Li, and Ying Shan. Taming rectified flow for inversion and editing. In *International Conference on Machine Learning*, 2025. [7](#), [8](#)
- [57] Yuancheng Wang, Zeqian Ju, Xu Tan, Lei He, Zhizheng Wu, Jiang Bian, et al. Audit: Audio editing by following instructions with latent diffusion models. *Advances in Neural Information Processing Systems*, 36:71340–71357, 2023. [3](#)
- [58] Duncan Williams, Victoria J Hodge, Lina Gega, Damian Murphy, Peter I Cowling, and Anders Drachen. Ai and automatic music generation for mindfulness. In *2019 AES International Conference on Immersive and Interactive Audio: Creating the Next Dimension of Sound Experience*. York, 2019. [2](#)
- [59] Yuxuan Wu, Yifan He, Xinlu Liu, Yi Wang, and Roger B Dannenberg. Transplayer: Timbre style transfer with flexible timbre control. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE, 2023. [1](#)
- [60] Sihan Xu, Yidong Huang, Jiayi Pan, Ziqiao Ma, and Joyce Chai. Inversion-free image editing with language-guided diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9452–9461, 2024. [2](#), [3](#)
- [61] Dongchao Yang, Jianwei Yu, Helin Wang, Wen Wang, Chao Weng, Yuexian Zou, and Dong Yu. Diffsound: Discrete diffusion model for text-to-sound generation. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 31:1720–1733, 2023. [3](#)
- [62] Li-Chia Yang, Szu-Yu Chou, and Yi-Hsuan Yang. Midinet: A convolutional generative adversarial network for symbolic-domain music generation. *arXiv preprint arXiv:1703.10847*, 2017. [2](#)
- [63] Ruibin Yuan, Hanfeng Lin, Shuyue Guo, Ge Zhang, Jiahao Pan, Yongyi Zang, Haohe Liu, Yiming Liang, Wenye Ma, Xingjian Du, et al. Yue: Scaling open foundation models for long-form music generation. *arXiv preprint arXiv:2503.08638*, 2025. [7](#)
- [64] Sicheng Zhao, Yaxian Li, Xingxu Yao, Weizhi Nie, Pengfei Xu, Jufeng Yang, and Kurt Keutzer. Emotion-based end-to-end matching between image and music in valence-arousal space. In *Proceedings of the 28th ACM International Conference on Multimedia*, pages 2945–2954, 2020. [2](#)
- [65] Le Zhuo, Ruoyi Du, Han Xiao, Yangguang Li, Dongyang Liu, Rongjie Huang, Wenze Liu, Xiangyang Zhu, Fu-Yun Wang, Zhanyu Ma, et al. Lumina-next: Making lumina-t2x stronger and faster with next-dit. *Advances in Neural Information Processing Systems*, 37:131278–131315, 2024. [4](#), [6](#)